

Total No. of Questions : 10]

P3266

SEAT No. :

[Total No. of Pages : 4

[5670]-534

B.E. (Mechanical Engineering)
FINITE ELEMENT ANALYSIS

(2015 Pattern) (End Sem) (Semester - I) (402044A) (Elective - I)

Time : 2½ Hours]

[Max. Marks : 70

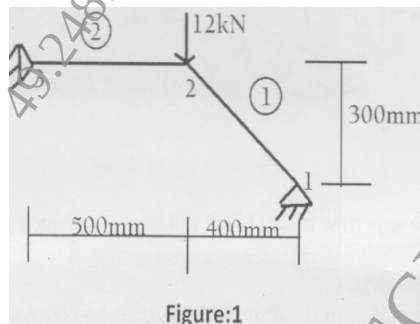
Instructions to the candidates:

- 1) *Draw suitable neat diagram, wherever necessary.*
- 2) *Figures to the right indicate full marks.*
- 3) *Use of electronic pocket calculator is allowed.*
- 4) *Assume suitable data, if required.*

- Q1)** a) Explain in brief Rayleigh-Ritz method and relation between FEM and Rayleigh-Ritz method **[6]**
- b) List different types of analysis and explain thermal analysis. **[4]**

OR

- Q2)** For truss given in figure 1, determine: **[10]**
- a) Nodal displacements
 - b) Stresses in each element and
 - c) Reaction at support. Take $E = 200 \times 10^3 \text{ N/mm}^2$ and $A = 200 \text{ mm}^2$



- Q3)** a) Explain pascal's triangle with neat sketch show that pascal's triangle satisfies all convergence conditions required for deciding interpolation function formulation. **[6]**

P.T.O.

- b) Write element stiffness matrix for plane stress/strain problems and explain each terms of it. [4]

OR

- Q4)** a) The nodal coordinate of triangular element are as shown in figure 2. The x coordinate of interior point P is 3.3 and shape function $N_1 = 0.3$. Determine N_2 , N_3 and y-coordinate of point P. (Use co-factor method)[4]

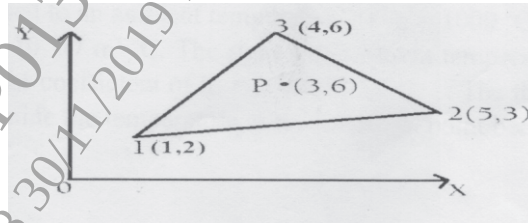


figure:2

- b) Write a short note on Element stiffness matrix and load vector for Beam element. [6]

- Q5)** a) Write short note on :

- Natural coordinates of element and its significance in FEA
- mesh refinements

- b) For the element shown in figure 3 assemble jacobian matrix for the Gauss point (0.7,0.5), also Check the element for uniqueness mapping. [8]

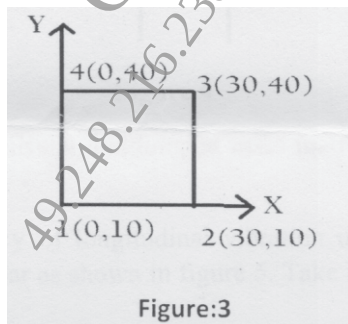


Figure:3

OR

- Q6)** a) Write a short note on jacobian matrix for “4-node Quadrilateral element” [8]

- b) Evaluate Integral : $I = \int_{y=4}^{y=6} \int_{x=-2}^{x=2} (1-x)^2 (4-y)^2 dx dy$ [8]

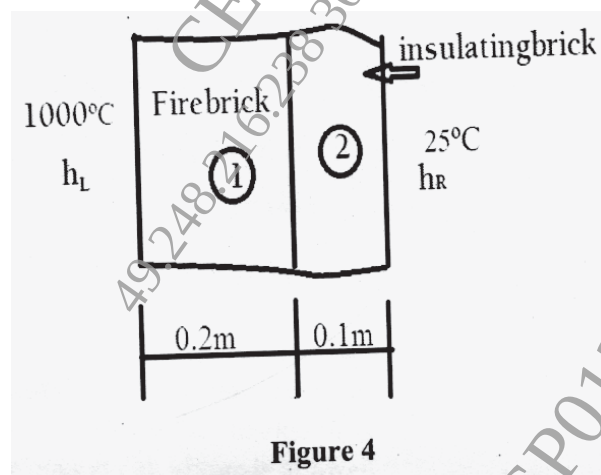
Q7) a) Write a note on “Heat transfer through pin-Fin” Explain with appropriate governing equation. [8]

b) A metallic fin, with thermal conductivity $70 \text{ W/m}^\circ\text{K}$, 1 cm radius and 5 cm long extends from a plane wall whose temperature is 413°K . Determine the temperature distribution along the fin if heat is transfer to ambient air at 293°K with heat transfer coefficient of $5 \text{ W/m}^2\text{K}$. Take two elements along the fin. [10]

OR

Q8) a) Write Element thermal conductivity(stiffness) matrix and Element thermal load vector. Compare them Element stiffness matrix and element load vector of Bar element respectively. [8]

b) A composite wall as shown in figure 4 is composed of two homogeneous slabs in contact. Let thermal conductivities be $K_1 = 1 \text{ W/m}^\circ\text{C}$ for firebrick slab 1 and $K_2 = 0.3 \text{ W/m}^\circ\text{C}$ for insulating slab 2. The left side is exposed to an ambient temperature of $T_{\text{fl}} = 1000^\circ\text{C}$ inside the furnace with heat transfer coefficient of $h_1 = 10 \text{ W/m}^2\text{C}$. The right side ambient temperature is $T_{\text{fr}} = 25^\circ\text{C}$ outside of the furnace with heat transfer coefficient of $h_R = 3 \text{ W/m}^2\text{C}$. The thickness of the slabs are $L_1 = 0.2 \text{ m}$ and $L_2 = 0.1 \text{ m}$. Determine the temperature at the left edge, point between the two slabs and right of the composite wall. [10]



Q9) a) List difference between consistent and lumped mass matrix technique for modal analysis of structure. [10]

- b) Find the natural frequency of longitudinal vibration using consistent and lumped mass matrix method with one element of bar as shown in figure 5. Take $E = 2 \times 10^{11} \text{ N/m}^2$, $\rho = 7800 \text{ kg/m}^3$, $L = 1 \text{ m}$ [6]

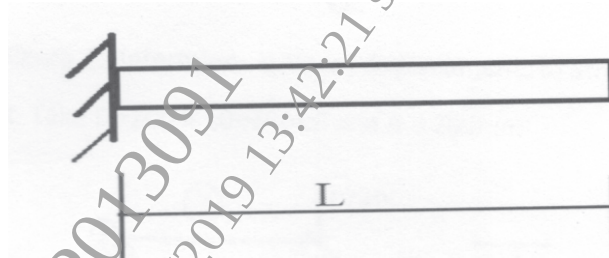


Figure:5

- Q10) a) Find the natural frequency of longitudinal vibration of the stepped bar of cross-sectional areas $A_1 = 2500 \text{ mm}^2$ and $A_2 = 1200 \text{ mm}^2$ having equal stepped length $L = 1 \text{ m}$ as shown in figure 6. (Use consistent mass matrix) [10]

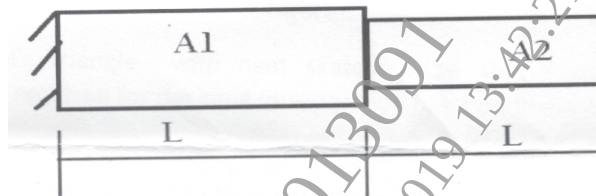


Figure:6

- b) Write a dynamic equation and explain each term. Convert this into a Eigen value problem and explain its significance. [6]

